

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2019
Math 2123
(Mathematics-III)

Full Marks: 210

Time: 3 hrs.

- N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section - A

1. (a) Define homogeneous system of linear equations. Find a cubic polynomial (13)
 $P(x) = a + bx + cx^2 + dx^3$ such that, $P(1) = 1$, $P'(1) = 5$, $P(-1) = 3$ and $P'(-1) = 1$.

- (b) What is meant by orthogonal matrix? Determine whether the matrix A is (10)
orthogonal or not.

$$A = \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ 2/3 & -1/3 & 2/3 \\ -1/3 & 2/3 & 2/3 \end{bmatrix}$$

- (c) Determine the values of 'a' for which the following system has no solutions, (12)
exactly one solution or infinitely many solutions.

$$\begin{aligned} x + 2y + z &= 2 \\ 2x - 2y + 3z &= 1 \\ x + 2y + (a^2 - 3) &= a \end{aligned}$$

2. (a) Define rank of the matrix. Find the rank of the matrix B, where k is a real (10)
number.

$$B = \begin{bmatrix} k & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-k \end{bmatrix}$$

- (b) Using elementary row operations, find the inverse of the matrix: (12)

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

- (c) If $M = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ then find all eigenvalues of M and the corresponding (13)
eigenvectors. Also evaluate an invertible matrix P such that $P^{-1}MP$ is
diagonal.

3. (a) What is symmetric matrix? Find all values of a, b and c for which matrix A is (10)
symmetric.

$$A = \begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$$

- (b) Define Laplace transform. Find Laplace transform of (12)
 $t^2 e^{-t} \cos t + (t-1)^2$

- (c) Define unit step function. Express the following function in terms of unit (13)
step. Hence find its Laplace transform.

$$f(t) = \begin{cases} 2 & 0 \leq t < 3 \\ 2t & 3 \leq t < 5 \\ \frac{1}{7}t^2 & t \geq 5 \end{cases}$$

4. (a) State convolution theorem for Inverse Laplace transform. Using partial (12)
fraction decompositions evaluate

$$L^{-1} \left\{ \frac{5s+3}{(s-1)(s^2+2s+5)} \right\}$$

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- (b) Using the concept of Laplace transform prove that

$$\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$$

- (c) Graph of a function $f(t)$ is given below. Find Laplace transform of $f(t)$.

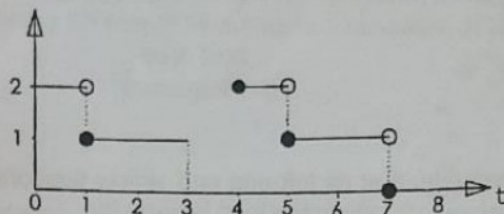


Figure – Question 4 (c)

Section – B

5. (a) The scalar triple product can be interpreted as the volume of a parallelepiped. Now you give the details about this interpretation and then find the volume of a parallelepiped determined by the vectors

$$2\hat{i} - 3\hat{j} + 4\hat{k}, \hat{i} + 2\hat{j} - \hat{k}, 3\hat{i} - \hat{j} + 2\hat{k}$$

- (b) Find the derivative of the function $f(x, y, z) = yz + zx + xy$ at the point $(1, -1, 2)$ in the direction of $\vec{A} = 3\hat{i} + 6\hat{j} - 2\hat{k}$.

- (c) Show that $\vec{F} = (e^x \cos y + yz)\hat{i} + (xy - e^x \sin y)\hat{j} + (xy + z)\hat{k}$ is conservative and find a potential function for it.

6. (a) Show that $\text{div}(\phi \vec{A}) = (\text{grad } \phi) \cdot \vec{A} + \phi \text{div } \vec{A}$ and verify for $\phi = e^{xyz}$ and $\vec{A} = ax\hat{i} - by\hat{j} + cz\hat{k}$.

- (b) If $\vec{A} = (y^2 + 2x)\hat{i} + (3y - 4x)\hat{j}$ evaluate $\oint_C \vec{A} \cdot d\vec{r}$ around the triangle c of the following figure in the indicated direction.

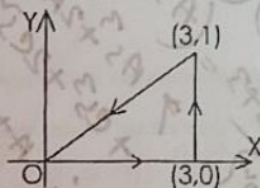


Figure: Question – 6 (b)

- (c) Evaluate

$$\iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} ds$$

where $\vec{F} = (x + 2y)\hat{i} - 3z\hat{j} + x\hat{k}$ and s is the surface of $2x + y + 2z = 6$ located in the first octant.

7. (a) Evaluate

$$\iiint_V (\vec{\nabla} \times \vec{F}) \cdot d\vec{v}$$

where V is the closed region bounded by the planes $x = 0, y = 0, z = 0$ and $2x + 2y + z = 4$.

- (b) Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.

- (c) For $\vec{A} = 2yz\hat{i} - x^2y\hat{j} + xz^2\hat{k}$ and $\phi = 2x^2yz^3$ find $(\vec{A} \times \vec{\nabla})\phi$ and $\vec{A} \times (\vec{\nabla}\phi)$.

2. (a) A beam is embedded at its left end and simply supported at its right end. (20)
Find the deflection $y(x)$ when the load is given by

$$w(x) = \begin{cases} w_0; & 0 < x < \frac{L}{2} \\ 0; & \frac{L}{2} \leq x < L \end{cases}$$

- (b) Using Laplace transform solve the differential equation

$$y'' + 2y' + 5y = e^{-t} \sin t$$
$$y(0) = 0 \quad y'(0) = 1$$

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$$\therefore 3Y - 5.0 - 1 + 2(Y - 0) + 5Y = \frac{1}{(4+1)+1} = \frac{1}{5}$$

$$Y - 5.0 - 1 + 2.5$$

$$Y = \frac{(5+2S+3)(5+2S+5)}{(5+2S+2)(5+2S+1)}$$

$$A + e = 0$$

$$\begin{aligned} A + C &= 0 \\ A + B + 2C + D &= 0 \\ 5A + 2B + 2C + D &= 0 \\ 5B + 2C + D &= 0 \\ A + B + 2C + D &= 0 \end{aligned}$$

$$\frac{s^2 + 2s + 3}{(s+1)^2} = \frac{s^2 + 2s + 1 + 2}{(s+1)^2} = \frac{(s+1)^2 + 2}{(s+1)^2} = 1 + \frac{2}{(s+1)^2}$$

$$\frac{(s+1)^2 + 2}{(s+1)^2 + 4}$$

$$y = \frac{(x+1)^3}{3} + \frac{x^2}{2} + \frac{(x+1)^2 + 2x}{2} + \frac{x}{3} \sin 2x + \frac{1}{3} \sin 2x$$